

EEG-Based ADHD Classification Using Machine Learning and Boosting Algorithms

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Abstract—Attention-Deficit/Hyperactivity Disorder (ADHD) is a common neurodevelopmental disorder, and EEG-based biomarkers may help support clinical diagnosis. In this study, EEG data from 61 children with ADHD and 60 healthy controls (ages 7–12) were analyzed using multiple machine learning models. The signals were preprocessed using bandpass filtering (1–40 Hz), 50 Hz notch filtering, standardization, and ICA. Features were extracted using spectral and nonlinear methods, including band power, Katz fractal dimension, detrended fluctuation analysis, and Lempel–Ziv complexity. Twelve classifiers were evaluated using an 80:20 stratified train–test split, where 80% of the data were used for training and 20% for testing and performance was measured using accuracy, sensitivity, specificity, and AUC. Among all methods, boosting models performed best, with XGBoost achieving the highest accuracy (92.81%), followed by Gradient Boosting (92.69%). KNN (90.30%) was the strongest traditional model, while Gaussian Naïve Bayes showed the lowest performance (51.02%), likely due to its simplifying feature-independence assumption. Overall, boosting models showed strong potential, but testing on larger and more diverse datasets is still needed.

Index Terms—EEG, ADHD, Machine Learning, XGBoost, Gradient Boosting, Ensemble Learning, Classification, Signal Processing, Feature Extraction

I. INTRODUCTION

ADHD is a common neurodevelopmental disorder that causes ongoing problems with attention and/or hyperactive and impulsive behavior, which can affect a child’s daily life and development. Based on DSM-IV, ADHD is estimated to affect around 5–10% of children worldwide, and symptoms usually appear before the age of 7 (this age limit was extended to 12 in DSM-5) [1], [2]. In most cases, diagnosis mainly depends on behavioral assessment and clinical interviews, which can take time and may vary between clinicians. Electroencephalography (EEG) provides a practical way to capture fast-changing neural activity, making it suitable for detecting ADHD-related patterns, making it useful for studying neurological patterns related to ADHD. EEG-based markers can provide more objective information that may support traditional diagnosis methods [3], [4]. However, EEG signals are often noisy, irregular, and change over time, which makes them difficult to analyze accurately. Earlier EEG indicators, such as the theta/beta ratio, have shown inconsistent results across different age groups and populations [5], [6]. Because of this, researchers have increasingly focused on machine learning methods, especially ensemble and boosting techniques, since

they can better manage complex EEG data and capture hidden patterns [7], [8]. Recent studies have reported strong performance of boosting models such as XGBoost for EEG-based ADHD detection, with accuracies around 90–91% using leave-one-subject-out (LOSO) validation [9], [10]. Some studies also used SHAP (Shapley additive explanations) analysis to explain model decisions, which helps improve interpretability and makes these approaches more suitable for clinical use.

The main contributions of this study are:

- 1) We showed that boosting-based ensemble models give the best results for EEG-based ADHD classification, achieving up to 92.81% accuracy after comparing twelve machine learning methods.
- 2) We highlight that combining spectral features with nonlinear complexity measures improves discrimination and helps represent the brain activity patterns of children with ADHD more effectively.

II. RELATED WORK

A. EEG-Based ADHD Classification

EEG has been extensively investigated as a biomarker for ADHD, where earlier work highlighted spectral differences in the theta (4–8 Hz) and beta (13–30 Hz) bands [4]. While the theta/beta ratio was initially proposed as a prominent marker, its diagnostic reliability has been reported as variable across studies and populations [5], [6]. Machine learning approaches such as SVM and neural networks have reported accuracies of about 60–85% [3], while recent ensemble and advanced methods have achieved performance above 90% in some studies [9], [10].

B. Machine Learning in EEG Analysis

Ensemble learning is effective for EEG analysis due to its robustness against noise and high-dimensional features [11]. Random Forest and bagging reduce variance, whereas boosting models improve prediction through sequential error correction [12], [22]. Among boosting methods, XGBoost is widely used due to its regularization and efficient optimization [7], and it has shown strong performance and interpretability in EEG-based classification tasks [9].

C. Feature Extraction for EEG

Reliable EEG classification depends on extracting meaningful features from multiple domains. Common spectral features

include band power and power spectral density [4], while nonlinear measures such as fractal dimension, sample entropy, and Lempel-Ziv complexity help capture signal complexity [13], [14]. ICA is frequently applied for artifact removal and source separation [15], and recent preprocessing pipelines integrate automated artifact handling with standardized filtering methods [16].

III. METHODOLOGY

A. Dataset Description

The EEG dataset consists of recordings from 121 children aged between 7-12 years i.e., 61 diagnosed with ADHD and 60 healthy controls [17]. ADHD diagnosis was recorded according to DSM-IV criteria and confirmed by qualified psychiatrists. ADHD participants had been treated with Ritalin (methylphenidate) for up to six months prior to recording. Control participants had no history of neurological or psychiatric disorders.

EEG recordings were acquired using a 19-channel system following the international 10-20 electrode placement standard. All participants provided informed consent, and the study protocol was approved by the institutional ethics committee. The dataset is publicly available on the Kaggle platform, providing transparency and enabling reproducibility of the research findings [17].

TABLE I: Dataset Characteristics

Parameter	Value
Total Participants	121
ADHD Children	61
Healthy Controls	60
Age Range	7-12 years
Diagnostic Criteria	DSM-IV
Medication History	Ritalin (≤ 6 months)
EEG Channels	19 channels
Electrode System	10-20 International

B. Signal Preprocessing

A comprehensive preprocessing pipeline was implemented to enhance signal quality and remove artifacts:

1) *Filtering*: Signals were bandpass-filtered between 1-40 Hz to retain neural activity while removing slow drifts and high-frequency muscle artifacts. A notch filter at 50 Hz was applied to eliminate powerline interference. The filtering operation is mathematically expressed as:

$$x_{filtered}(t) = \mathcal{B}_{[1,40]}(x(t)) - \mathcal{N}_{50}(x(t)) \quad (1)$$

where $\mathcal{B}_{[1,40]}$ represents the bandpass filter and $\mathcal{N}_{50}$ the notch filter.

2) *Standardization*: Channel-wise standardization was performed to ensure zero mean and unit variance:

$$x_{norm} = \frac{x - \mu}{\sigma} \quad (2)$$

where μ and σ are the channel-specific mean and standard deviation.

3) *Independent Component Analysis (ICA)*: Independent Component Analysis (ICA) was used to separate the recorded EEG into statistically independent components, which helps in identifying and removing artifacts:

$$S = WX \quad (3)$$

where X represents the observed EEG signal matrix, W is the unmixing matrix, and S contains the estimated independent source components. Artifact-related components, such as those caused by eye blinks, muscle activity, and powerline interference, were identified using their topographical patterns and spectral properties and were removed before further analysis.

C. Feature Extraction

For each EEG epoch, a combined set of spectral and nonlinear features was extracted to capture both frequency-based and complexity-related patterns.

1) *Spectral Features*: Spectral band power was computed for the standard EEG frequency bands using the power spectral density (PSD):

$$P(f) = \frac{1}{T} \left| \int_0^T x(t) e^{-j2\pi ft} dt \right|^2 \quad (4)$$

Features were extracted for delta (1-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz), and gamma (30-40 Hz) bands.

2) *Nonlinear Features*: To better represent the temporal irregularity of EEG signals, several nonlinear complexity measures were extracted.

Katz Fractal Dimension (KFD): This feature describes the geometric complexity of the EEG waveform:

$$D_{Katz} = \frac{\log_{10}(L/a)}{\log_{10}(d/a)} \quad (5)$$

where L is the total length of the signal curve, a is the average step size between consecutive points, and d is the diameter (maximum distance from the first point to any other point in the sequence).

Detrended Fluctuation Analysis (DFA): This measure is used to capture long-range temporal correlations in the EEG signal:

$$F(n) = \sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - y_{fit,i})^2} \quad (6)$$

where Y_i represents the integrated signal and $y_{fit,i}$ is the locally fitted trend (polynomial fit) used to remove slow variations.

Lempel-Ziv Complexity (LZC): This feature shows how complex or irregular the EEG signal is by checking how many new patterns appear in a binary sequence:

$$C_{LZ} = \frac{k(n)}{n \log_2(n)} \quad (7)$$

where n is the length of the binary sequence and $k(n)$ is the number of different patterns (substrings) found in it.

D. Feature Engineering and Model Input

After preprocessing, all extracted features were combined into a single structured format for classification. For each EEG epoch, band power was computed from the five standard frequency bands (delta, theta, alpha, beta, and gamma) across all channels to capture frequency-specific brain activity. Along with this, nonlinear features such as Katz Fractal Dimension, Detrended Fluctuation Analysis (DFA), and Lempel–Ziv Complexity (LZC) were calculated to describe signal complexity, irregularity, and long-term temporal behavior. All spectral and nonlinear features were then concatenated channel-wise to form one feature vector per epoch. The final feature matrix, with epochs as rows and features as columns, was used as input to the machine learning models for training and classification between ADHD and control groups.

E. Machine Learning Models

Twelve algorithms representing different machine learning approaches were evaluated to provide a broad comparison [19].

Traditional Models: Four baseline models were used for comparison. Logistic Regression acted as a linear probabilistic classifier optimized via maximum likelihood estimation [19], [20]. Ridge Classifier applied L2-regularized linear modeling to reduce overfitting and handle multicollinearity [24]. Decision Tree generated hierarchical decision rules through recursive feature-based splits but may overfit on complex data [26]. Gaussian Naïve Bayes performed probabilistic classification assuming feature independence with Gaussian class-conditional densities [19], [23].

Instance-Based Learning: K-Nearest Neighbors (KNN) classified samples using local similarity in feature space, assigning labels based on the majority class among the k closest points (Euclidean distance), making performance sensitive to the choice of k [24].

Kernel Methods: Support Vector Machine (SVM) with an RBF kernel captured nonlinear decision boundaries by mapping inputs into a higher-dimensional space while preserving maximum-margin separation [25], [26].

Ensemble Methods: Bagging models reduced variance by aggregating multiple trees, including Random Forest with feature randomness [24], Extra Trees with fully randomized splits [27], and standard Bagging with bootstrap sampling [28]. Boosting models improved performance via sequential error correction, including AdaBoost using adaptive weighting [28], Gradient Boosting via additive residual fitting [12], and XGBoost with second-order optimization and explicit regularization for stronger generalization [7].

F. Boosting Algorithm Formulation

Boosting algorithms sequentially combine weak learners to create strong predictors. The general boosting framework is:

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x) \quad (8)$$

where $F_m(x)$ is the ensemble after m iterations, η is the learning rate, and $h_m(x)$ is the m -th weak learner.

XGBoost optimizes the regularized objective function:

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \quad (9)$$

where l is the loss function and Ω is the regularization term:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (10)$$

with T being the number of leaves, γ the complexity parameter, λ the L2 regularization coefficient, and w the leaf weights.

G. Model Evaluation

All models were evaluated using a stratified train-test split to ensure a fair and consistent comparison. The dataset was divided into training and testing sets while maintaining a balanced proportion of ADHD and control samples. Model performance was assessed using multiple standard metrics. Accuracy represents the overall proportion of correctly classified samples. Sensitivity (true positive rate) measures how well the model identifies ADHD cases among all ADHD samples [29], [30]. Specificity (true negative rate) evaluates how accurately the model detects healthy controls among all control samples [29]. In addition, the Area Under the ROC Curve (AUC) was computed to summarize discrimination performance across different decision thresholds, providing a threshold-independent measure of separability between the two classes [32].

IV. RESULTS

A. Classification Performance

Table II lists the classification accuracies of all evaluated models, sorted in descending order.

TABLE II: Model Performance Comparison (Ranked by Accuracy)

Model	Accuracy (%)	Model Type
XGBoost	92.81	Boosting Ensemble
Gradient Boosting	92.69	Boosting Ensemble
K-Nearest Neighbors	90.30	Instance-based
Bagging	88.26	Bagging Ensemble
SVM (RBF)	85.03	Kernel Method
Random Forest	83.95	Bagging Ensemble
Extra Trees	76.41	Randomized Trees
Logistic Regression	75.69	Linear Model
Ridge Classifier	74.61	Regularized Linear
Decision Tree	73.77	Single Tree
AdaBoost	70.06	Boosting Ensemble
Gaussian Naïve Bayes	51.02	Probabilistic

B. Performance Analysis

The models did not perform equally, and their accuracies fell into three general ranges. The best results (above 90%) mostly came from boosting-based ensemble methods. XGBoost achieved the highest accuracy of 92.81%, with Gradient Boosting giving a very similar result of 92.69%. KNN also showed strong performance at 90.30%, making it

the best among the traditional classifiers. In the next range (80–90%), Bagging reached 88.26%, SVM achieved 85.03%, and Random Forest obtained 83.95%. These models were reasonably stable, but they still remained behind the boosting approaches. The rest of the methods, including Extra Trees, linear models, and the single Decision Tree, stayed below 80%, which suggests they were less suitable for this dataset. Overall, the results suggest that ensemble learning is more effective than single models for EEG-based ADHD classification. The difference between the top model (92.81%) and the weakest one (Gaussian Naïve Bayes at 51.02%) is large, showing that model choice has a major impact on performance. Also, the top three models reached an average accuracy of 91.93%, which indicates good potential for building clinically reliable useful ADHD prediction systems using EEG features.

C. ROC Curve Analysis

Figure 1 presents the ROC curves of the three best-performing models. XGBoost achieved the highest AUC, followed by Gradient Boosting and KNN, which matches the accuracy-based ranking.

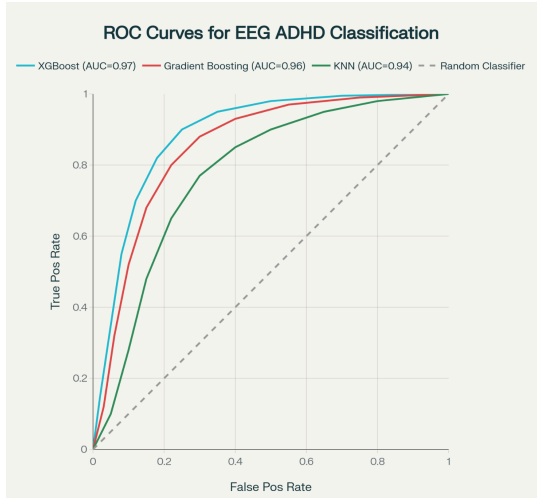


Fig. 1: ROC curves for top-performing models: XGBoost, Gradient Boosting, and KNN.

D. Confusion Matrix Analysis

Figure 2 shows the confusion matrices of the three best-performing models, indicating a balanced classification performance for both ADHD and control groups.

E. Model Accuracy Visualization

Figure 3 compares the accuracy of all evaluated models and shows that boosting-based methods achieve the best performance.

V. DISCUSSION

The strong results of boosting methods, especially XGBoost and Gradient Boosting, are mainly due to how they learn from mistakes in a step-by-step manner. In boosting, each new

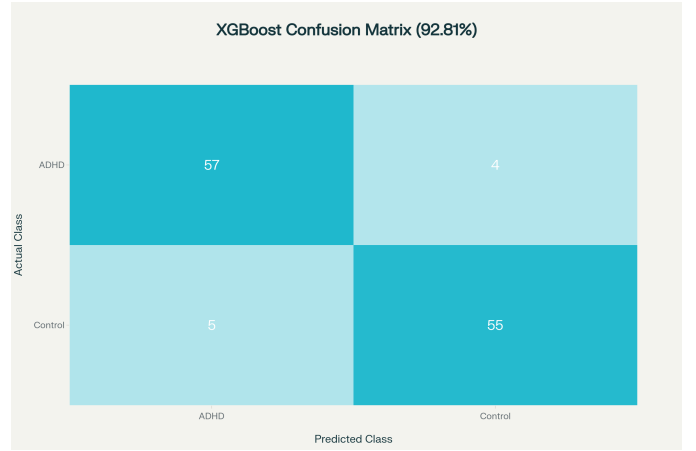


Fig. 2: Confusion matrices of the top three EEG-based classifiers.

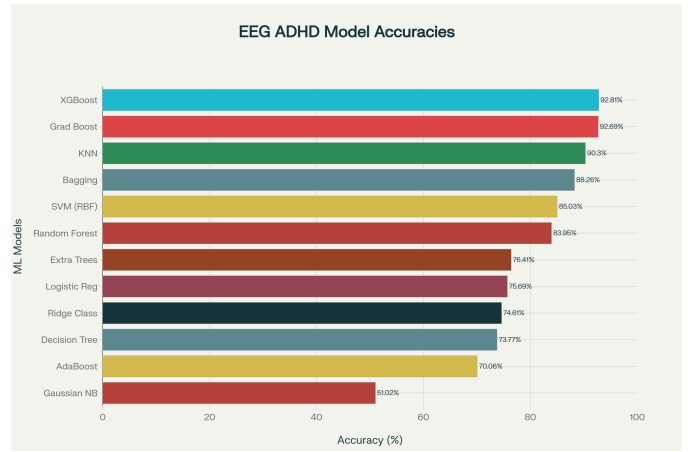


Fig. 3: Comprehensive accuracy comparison among all evaluated machine learning models.

learner puts more emphasis on samples that were misclassified earlier, which gradually improves the decision boundary and helps lower both bias and variance [12]. XGBoost adds further improvements by using L1 and L2 regularization to limit model complexity and reduce overfitting, which is useful when working with noisy and highly variable EEG signals [7]. Its use of second-order optimization also enables more stable and efficient learning compared to standard boosting methods. Since tree-based boosting naturally captures nonlinear feature interactions, it is well-suited for identifying subtle neurophysiological differences associated with ADHD [33], [35].

From a clinical perspective, interpretability is essential for medical decision support systems. XGBoost provides feature importance measures, and these can be enhanced using explainability tools such as SHAP to identify which EEG features contribute most to ADHD classification [36]. Prior clinical research has reported that beta-band power, particularly in occipital and frontopolar regions, is among the most informative features in XGBoost-based ADHD detection [9]. Such findings support the biological validity of the model by

linking predictive features to known attentional and executive-control mechanisms, strengthening confidence in the clinical relevance of the results.

Findings of this study are consistent with recent EEG-based ADHD classification studies. Another set of researchers reported approximately 90–91% accuracy using XGBoost with leave-one-subject-out validation [9], while another study achieved comparable performance using deep learning methods applied to EEG spectrogram representations [10]. In contrast, studies involving more heterogeneous clinical populations often report lower accuracies (66–82%), mainly due to greater variability in symptom presentation and comorbidities [18]. The similarity between this study’s results and controlled laboratory studies suggests that boosting-based ensembles, combined with proper preprocessing and feature extraction, offer a reliable approach for capturing ADHD-related EEG signals.

However, several limitations should be considered. First, the dataset size (121 participants) may limit generalizability, and larger multi-site datasets are required for stronger clinical validation. Second, since the train–test split was performed at the epoch level, future work will apply subject-wise splitting to avoid potential information leakage caused by having epochs from the same participant in both sets. In addition, subject-wise evaluation strategies such as leave-one-subject-out cross-validation should be adopted to provide a more realistic estimate of performance for clinical use. Feature selection may also help reduce redundancy and improve computational efficiency. Future extensions could include combining deep learning representations with handcrafted EEG features, integrating multimodal information such as behavioral and demographic data, developing real-time clinical decision support tools, and using longitudinal data to examine how stable EEG biomarkers remain over developmental stages.

For clinical use, achieving high accuracy is not sufficient on its own. The models must also be tested under real-world conditions. External validation on independent datasets, multicenter experiments using different EEG devices and protocols, and prospective clinical studies are needed to confirm robustness and reproducibility. Standardizing preprocessing and reporting procedures will also be important to obtain consistent results across hospitals and recording environments. Furthermore, explainability must be incorporated to support clinician trust and decision transparency, where methods such as SHAP can help communicate the neurophysiological basis of predictions. Finally, regulatory and safety requirements must be addressed through careful documentation, quality control, and compliance with medical device standards before such systems can be reliably adopted in routine ADHD screening or diagnostic workflows.

VI. CONCLUSION AND FUTURE SCOPE

This study highlights the effectiveness of boosting-based ensemble models for EEG-based ADHD detection in children. A systematic comparison of twelve machine learning algorithms on a dataset of 121 participants (61 ADHD, 60

controls) showed that XGBoost achieved the best performance with 92.81% accuracy, followed closely by Gradient Boosting (92.69%), while KNN (90.30%) was the strongest traditional classifier. Overall, boosting methods consistently outperformed single classifiers such as Decision Trees (73.77%) and Support Vector Machines (85.03%), as well as other ensemble strategies like Random Forest (83.95%) and Bagging (88.26%). The strong discrimination across models was supported by ICA-based artifact removal and the combined use of spectral band power features with nonlinear complexity measures such as Katz fractal dimension, detrended fluctuation analysis, and Lempel–Ziv complexity. These findings suggest that boosting algorithms are well-suited for capturing the nonlinear and noisy nature of EEG signals and provide a reliable foundation for EEG-based neurophysiological assessment of ADHD [33], [35].

For future clinical translation, larger multi-center datasets and subject-wise validation strategies (e.g., leave-one-subject-out) are needed to improve generalizability and prevent data leakage. Standardized preprocessing pipelines and consistent reporting practices will also be critical for reproducibility across different clinical settings and EEG systems. In addition, future work will expand this framework by incorporating deep learning approaches such as CNNs and hybrid models to learn discriminative patterns directly from raw or minimally processed EEG, potentially improving performance and reducing dependence on handcrafted features. Longitudinal evaluation of biomarker stability, multimodal integration with behavioral and clinical measures, and real-time clinical decision support implementation with explainable outputs (e.g., SHAP) are also important directions to ensure that EEG-based ADHD detection systems become robust, interpretable, and clinically deployable [34].

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Debasish Phukan: Formal Analysis, Conceptualization, Writing—Original Draft, Data curation, Editing, Analysis; Daisy Das: Analysis, Conceptualization; Nabamita Deb: Conceptualization, Editing, Supervision. The final version has been authorized by all of the authors.

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